

Andreas Panagopoulos

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EDUCATION

University of Patras

Patras, Greece

MEng in Electrical and Computer Engineering (GPA: 7.2/10)

Oct. 2022 – Aug. 2027 (Expected)

- Specialization: Computers (Software and Hardware).

Politecnico di Milano

Milan, Italy

Erasmus+ Exchange – MSc in Computer Science & Engineering

Feb. 2026 – July 2026

- Specialization: Deep Learning and Edge Intelligence.

National Center for Scientific Research (NCSR) “Demokritos”

Athens, Greece

Summer School Participant

July 2024

- Engaged in intensive lectures and hands-on workshops covering Ethics, AI, Physics, and Chemistry.
- Gained direct exposure to advanced laboratory instruments, including SEM and NMR Spectrometers.

General Lyceum (Geniko Lykeio)

Zakynthos, Greece

High School Diploma (Grade: 19.7/20)

Graduated June 2022

- Academic focus on Mathematics, Physics, and Chemistry.

WORK EXPERIENCE

Hellas Direct

Athens, Greece

Machine Learning Intern

Oct. 2026 – Dec. 2026 (Ongoing)

- Conducting applied research on Tabular Foundation Models (TabFMs) for automated risk assessment and insurance underwriting.
- Investigating deep learning architectures for pricing algorithms, benchmarking performance against traditional gradient-boosted tree baselines.
- Exploring JEV-based models for making routing decisions between automated pricing or human underwriters

Hospitality Services

Various Locations, Greece

Server / Bartender / Assistant Server

Summers 2022 – 2025

- Delivered high-standard customer service in fast-paced seasonal hospitality and food service environments.

ACADEMIC & PERSONAL PROJECTS

Quadcopter Acrobatics | *Python, PyTorch, ESP32, Edge AI, Reinforcement Learning* August 2026 – October 2026

- Implemented a PPO-based policy for control of a quadcopter’s rotor speeds for acrobatics.
- Designed a novel architecture by combining relevant architectural decisions suggested by academic literature, showing impressive Sim2Real capabilities.

Edge AI: Indoor Room Localization | *Python, TensorFlow, ESP32-S3, Edge AI* May 2026 – June 2026

- Surveyed embedded and edge artificial intelligence methods applied to radar signal localization.
- Architected a lightweight U-Net-based neural network model, trained and post-quantized to INT8.
- Built an end-to-end processing pipeline encompassing raw radar pre-processing, inference, and post-processing.
- Benchmarked and validated real-time memory and processing constraints on an ESP32-S3 microcontroller.

Industrial Anomaly Detection | *Python, PyTorch, Jupyter, Colab* May 2026

- Reproduced state-of-the-art anomaly detection benchmarks on non-conventional industrial visual datasets.
- Designed and implemented an algorithm combining PatchCore and Dinomaly architectures to improve localization accuracy.

3D Video Game & Rendering Engine | *C++, OpenGL, GLSL, GPU* Feb. 2026

- Engineered a modular rendering pipeline from scratch utilizing OpenGL and modern GLSL programmable shaders.
- Implemented procedural terrain generation, realistic dynamic lighting models, and multi-layered texturing.
- Integrated custom physics-based modeling algorithms for realistic collision detection and character mechanics.

Full-Stack Basketball Management Platform | *Python, Flask, MySQL, TiDB Cloud, Render* Dec. 2025

- Elicited domain requirements and designed a normalized relational database schema using E-R modeling.
- Deployed a cloud-hosted MySQL database on TiDB Cloud and optimized complex queries for sub-500ms response times.
- Built and deployed a responsive Flask web application wrapper hosted on Render.

Bare-Metal DSP Audio Processing | *C, Assembly, Texas Instruments DSP* Aug. 2025

- Implemented real-time digital audio filtering algorithms in C and Assembly directly on bare-metal hardware.
- Resolved strict low-level constraints including audio clipping, buffer overflows, and fixed-point arithmetic precision.

Linux Kernel Device Driver | *C, Linux Kernel, QEMU, VirtualBox* Jan. 2025

- Developed a low-level Linux character device driver for keyboard hardware input handling.
- Diagnosed and resolved custom kernel bootloader integration issues inside a virtualized QEMU environment.

European Electricity Market Data Analysis | *Python, BeautifulSoup, Statistics* June 2023

- Automated the extraction of public energy dataset feeds directly from official European Union data portals.
- Conducted statistical analysis on EU market trends and delivered an analytical report on the Greek energy sector.

TECHNICAL SKILLS

Languages: C, C++, Python, CUDA, Verilog, Java, Assembly, SQL (MySQL), GLSL
Frameworks & Libraries: PyTorch, TensorFlow, Flask, NumPy, pandas, Matplotlib
Tools & Platforms: Linux, Git, Docker, QEMU, VirtualBox, TiDB Cloud, Render, MongoDB, MQTT, Ollama, MATLAB, L^AT_EX
Spoken Languages: Greek (Native), English (C2), German (B2), Italian (B2)